

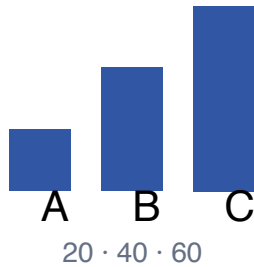
Choosing Effective Visual Encodings

From possible encodings to good encodings

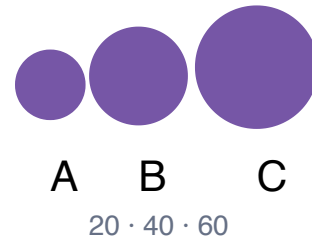
Information Visualization · Lecture 4

One attribute can be encoded in many ways

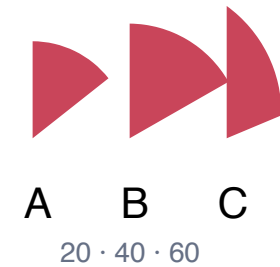
Length



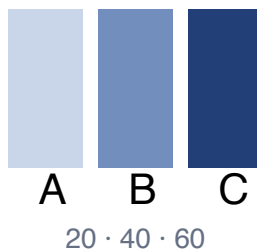
Area



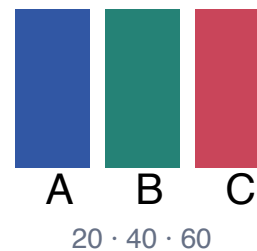
Angle



Luminance



Hue



**All five encodings
distinguish the three values.
Are they equally good?**

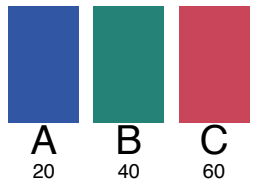
Agenda

-
- | | | |
|----|--|---|
| 01 | Expressiveness and effectiveness | Choose encodings that preserve meaning and support the task. |
| 02 | Discriminability and separability | Match channel capacity and limit interference. |
| 03 | Redundant encoding | Reinforce one attribute with multiple channels. |
| 04 | Weber's law | Explain relative comparison and pop-out through surrounding contrast. |
-

A good encoding must pass two tests

Expressiveness

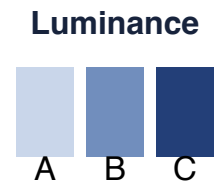
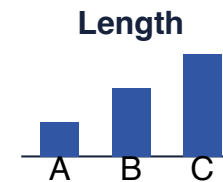
Does the encoding represent the meaning and relevant relationships of the data correctly?



The hues distinguish A, B, and C, but do they show that **20 < 40 < 60**?

Effectiveness

Does the encoding make the information important to the task easy and accurate to perceive?



Which encoding makes it easier to estimate **how much** A, B, and C differ?

A visualization should first be **expressive**, then as **effective as possible for the task**.

Expressiveness preserves the meaning of the data

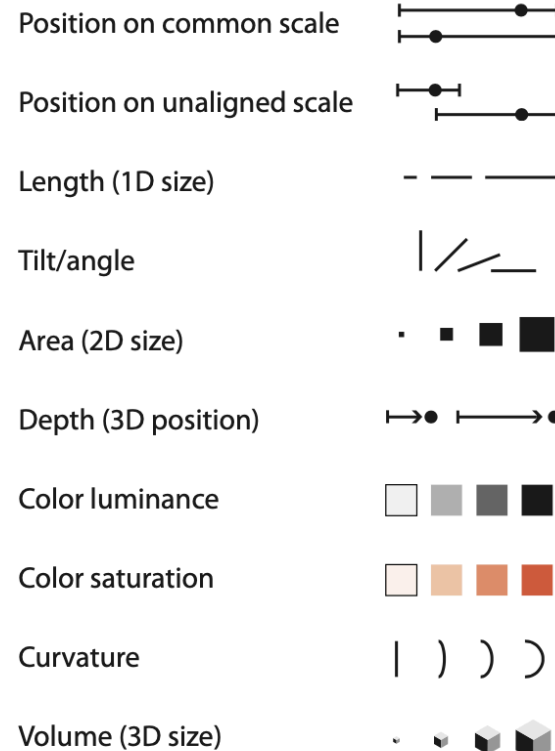
The expressiveness principle requires the meaning implied by a visual encoding to match the meaning of the data. The encoding must preserve the relationships needed for the task and must not suggest relationships that the data does not contain.

- **Categorical attributes** need channels that show identity or difference without implying order or magnitude.
- **Ordered attributes** need channels that viewers perceive as ordered.

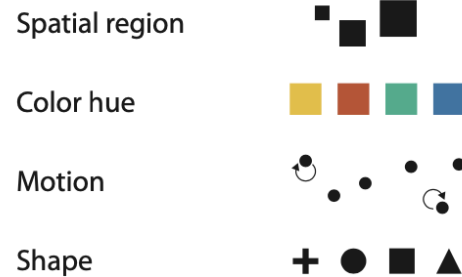
Design check: Does the visual imply the same relationships as the data?

Identity and magnitude channels support different judgements

➔ Magnitude Channels: **Ordered** Attributes



➔ Identity Channels: **Categorical** Attributes



What is it?

Identity channels

Primarily support judgments of **sameness and difference**.

Hue

Shape

Spatial region

How much?

Magnitude channels

Support judgments of **order or quantitative amount**.

Position

Length

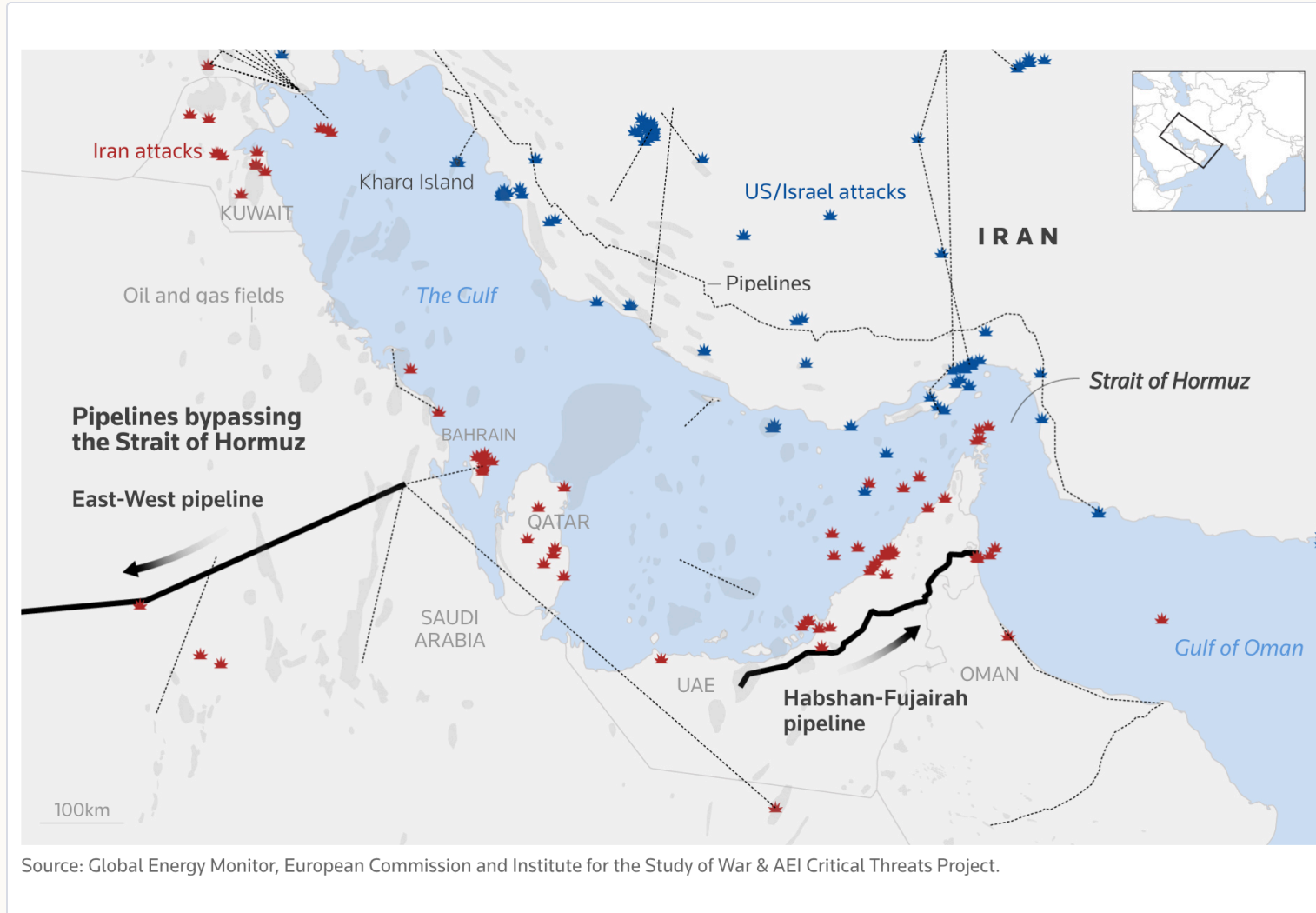
Angle

Area

Luminance

These are primary perceptual uses—not perfectly exclusive categories.

Categorical attributes need identity judgments



Iran, or, US → hue

The viewer needs to determine whether two points belong to the same or different categories.

No order or magnitude comparison is required.

Unordered hue cannot communicate quantitative order

Which is larger?

Is green higher than blue?
Hue provides difference, but
no natural magnitude order.

Different $\neq$ more or less

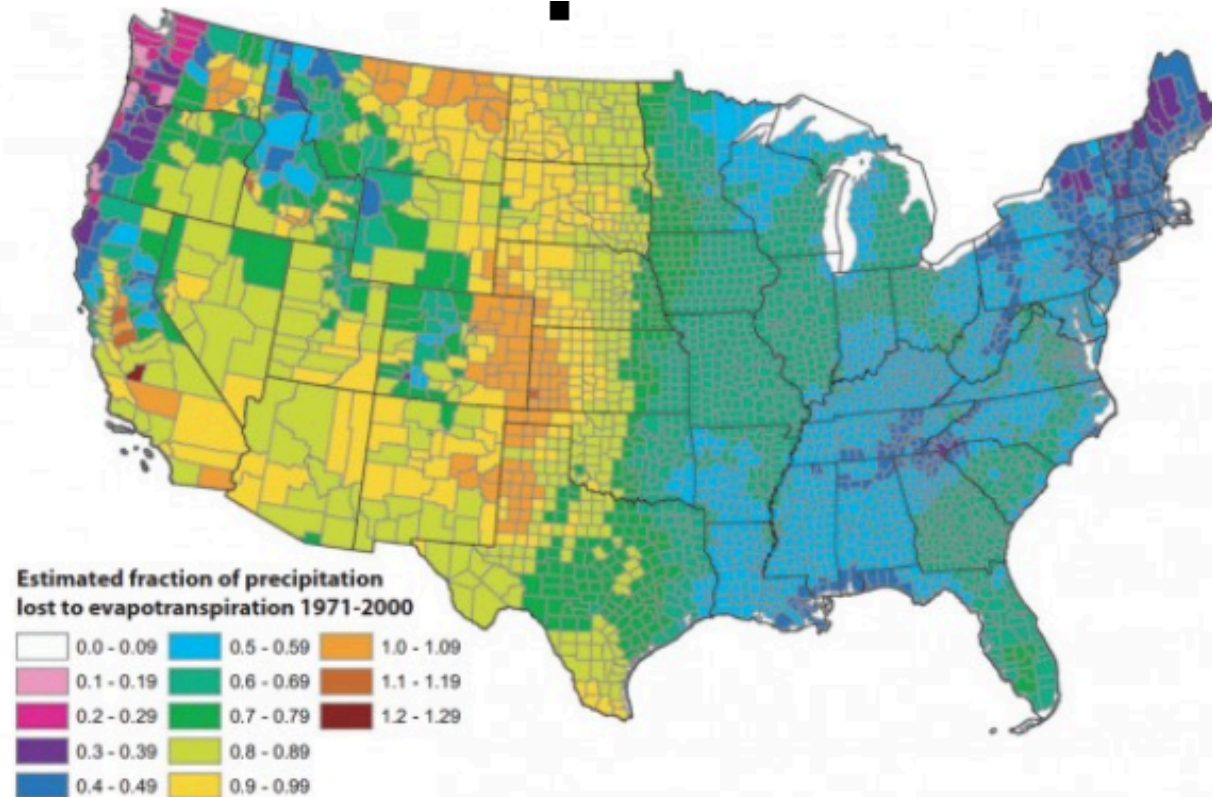
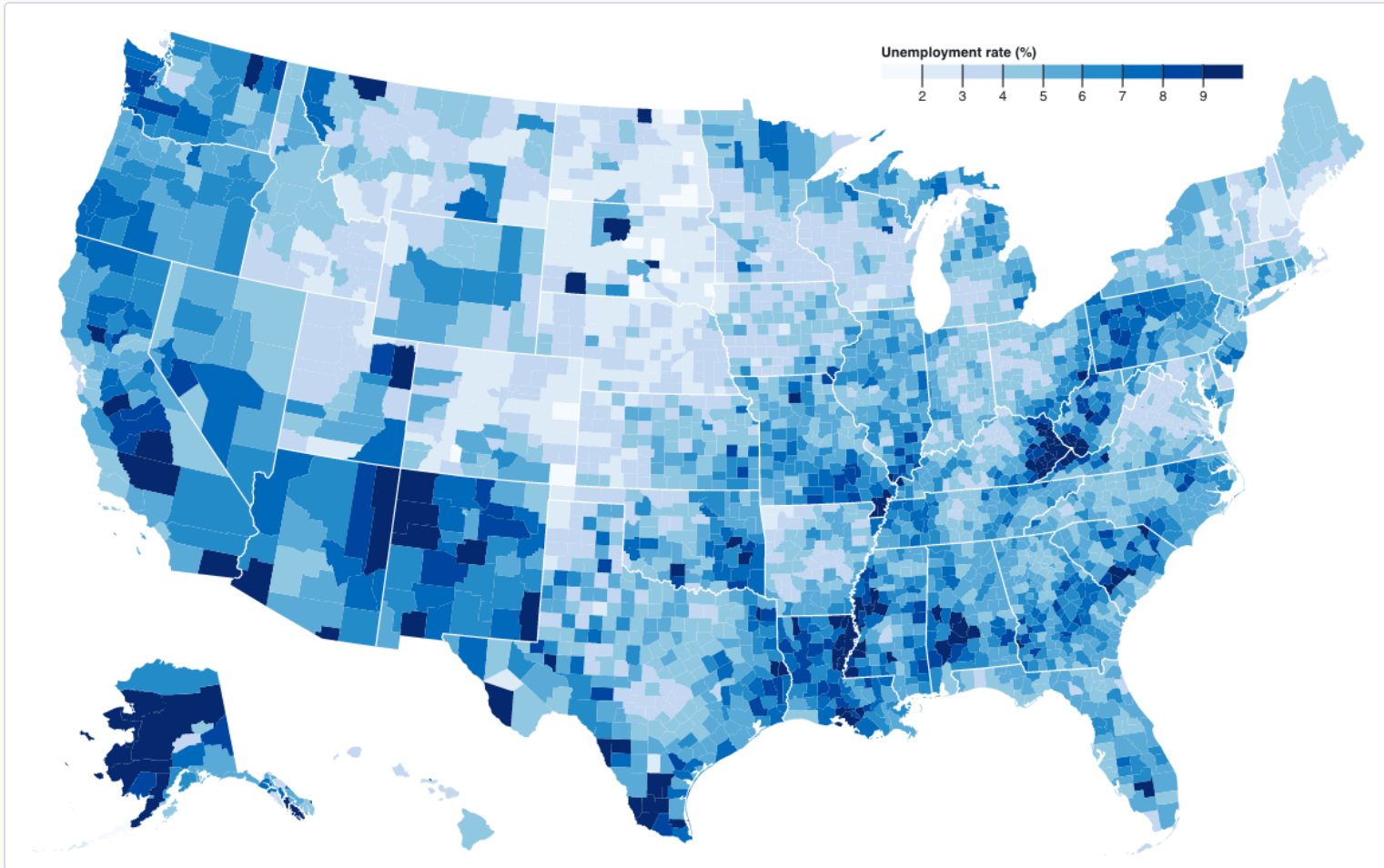


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

© 2000

Ordered lightness can communicate quantitative magnitude



Light → dark

A sequential color scale uses ordered lightness or luminance to represent increasing magnitude.

Viewers can recover a visual order even when exact values require the legend.

Size can invent an order that categories do not have

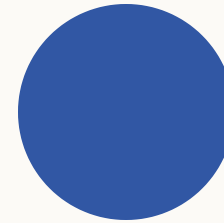
Task: Identify which company each mark represents.



Company A



Company B



Company C

What the data means

A, B, and C are categories. They have no inherent order or magnitude.

What size implies

Increasing circle size suggests an order such as $A < B < C$.

Design check failed: Size introduces a relationship that the data does not contain.

Which mappings are expressive?

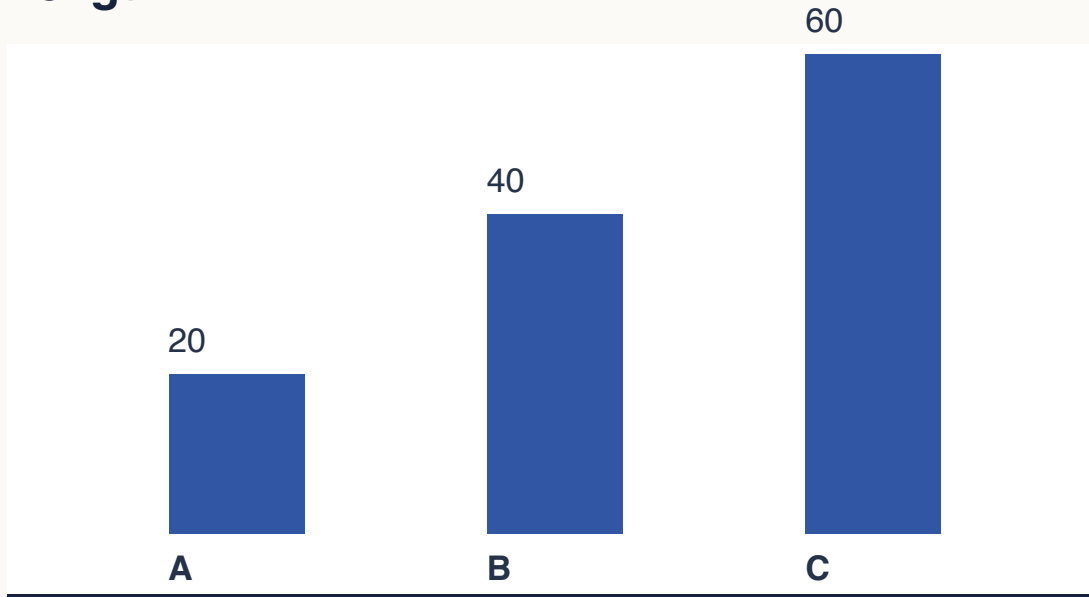
Attribute	Channel	Your judgment
Continent	Hue	Expressive / Problematic
Salary	Hue	Expressive / Problematic
Ranking	Vertical position	Expressive / Problematic
Country	Size	Expressive / Problematic
Temperature	Luminance	Expressive / Problematic
Product category	Shape	Expressive / Problematic

Explain your choice:

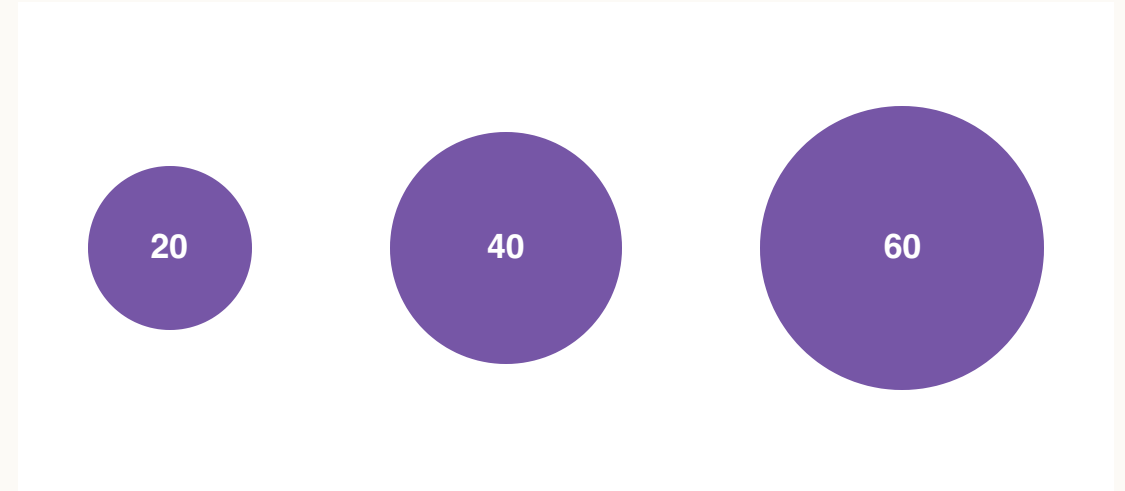
What relationship does the channel support, or falsely imply?

Two expressive encodings can differ in effectiveness

Length



Area



Both preserve magnitude. Which supports more accurate comparison?

Effectiveness assigns strong channels to what matters

The effectiveness principle requires the attributes most important to the visualization task to be encoded using the channels that support the most accurate and efficient perceptual judgments.

Task

What must the viewer judge?



Important attributes

Which data fields determine the answer?



Strong suitable channels

Which channels support that judgment?

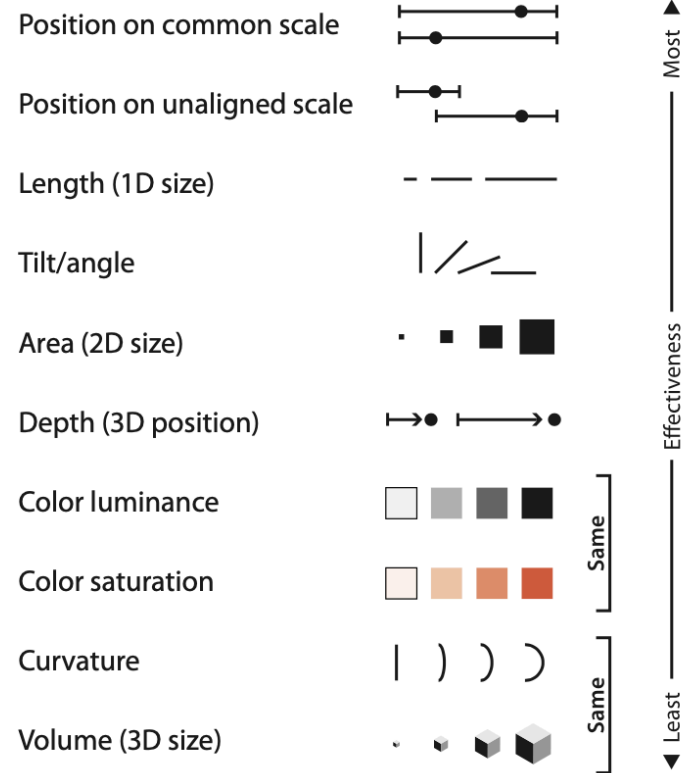
Perceptual accuracy provides evidence for effectiveness

Perceptual accuracy describes how closely a viewer's visual judgment corresponds to the underlying value or objective measurement represented by the graphic.

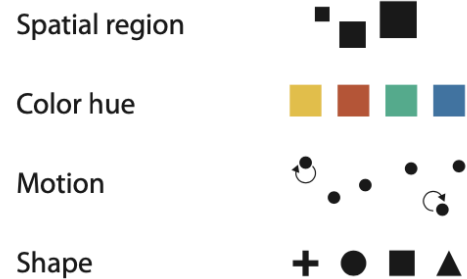
- Researchers compare estimation errors across visual channels.
- A channel that produces smaller errors for a judgment is generally more effective for that judgment.
- Accuracy is not the only design goal, but it is central when viewers must compare quantitative values.

Channels have ranks

➔ Magnitude Channels: Ordered Attributes



➔ Identity Channels: Categorical Attributes



How easily can categories be distinguished?

Ranking of identity channels

Spatial region > Hue > Motion > Shape

How accurately can values be compared?

Ranking of magnitude channels

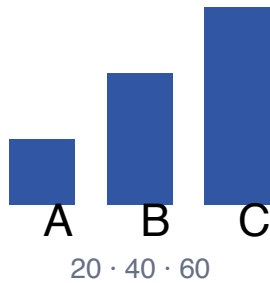
Selected channels:

Aligned position on a common scale > Unaligned position on identical scales > Length > Angle > Area > Luminance

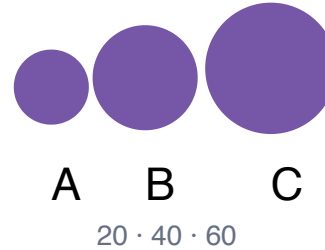
These rankings summarize general perceptual effectiveness; the best choice still depends on the task and context.

Higher channel rank makes it easier to compare

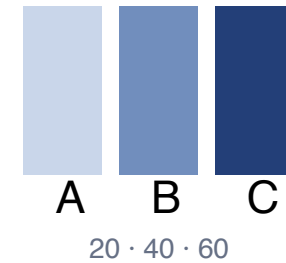
Length



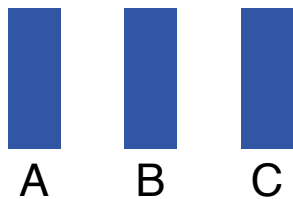
Area



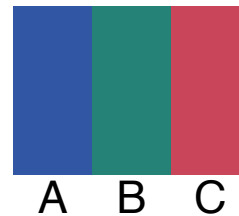
Luminance



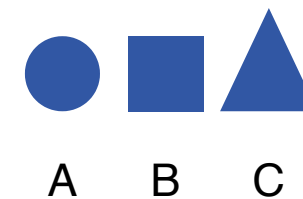
Spatial region



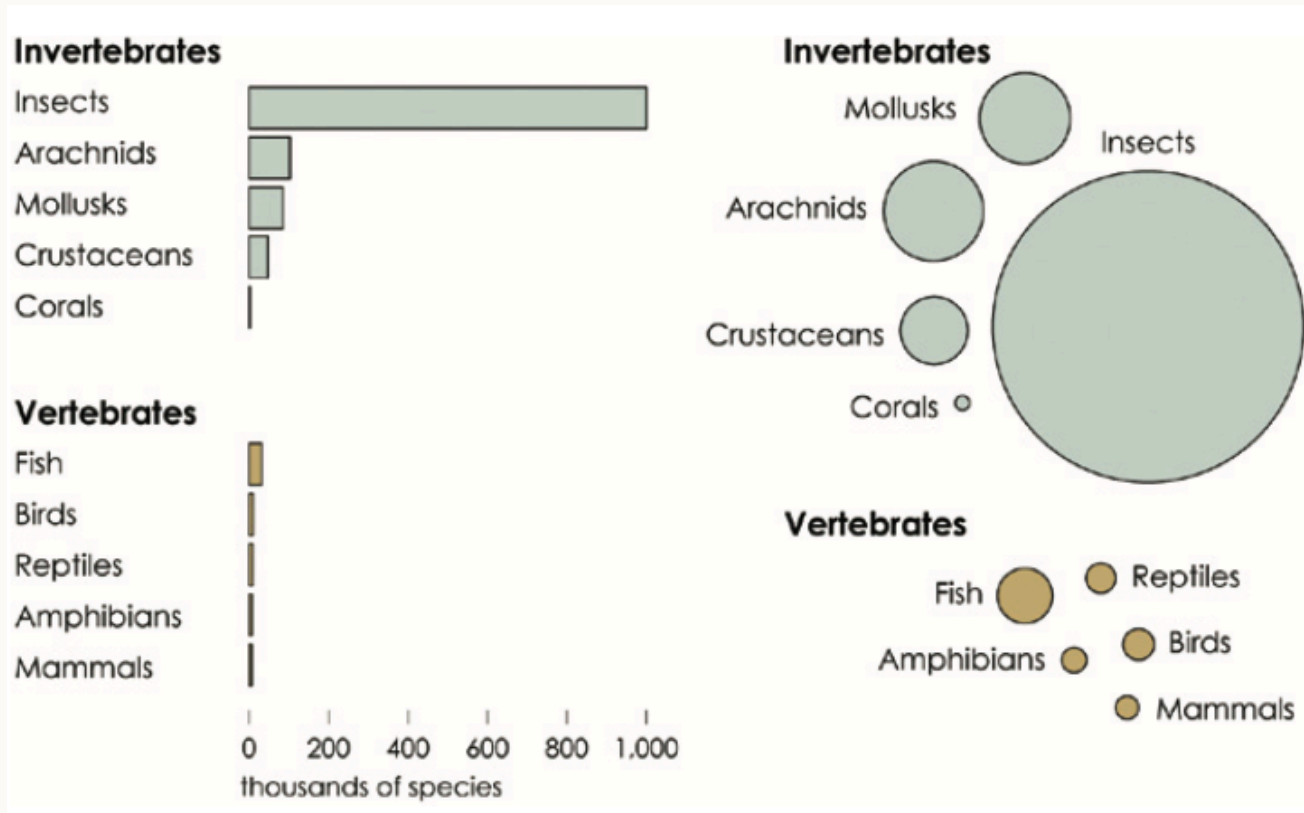
Hue



Shape



The best choice still depends on the task and context



Visualizing the numbers of species of animals

- The insect count is exponentially larger than the others.
- With **length** on a common scale, the remaining bars become too short to distinguish.
- **Size** mitigates this problem because circle diameters vary less dramatically.
- Although area ranks below length for precise comparison, it shows all groups more effectively in this context.

Effectiveness depends on the **accuracy** and **efficiency** of the perceptual judgment required by the task.

Discriminability limits distinguishable values

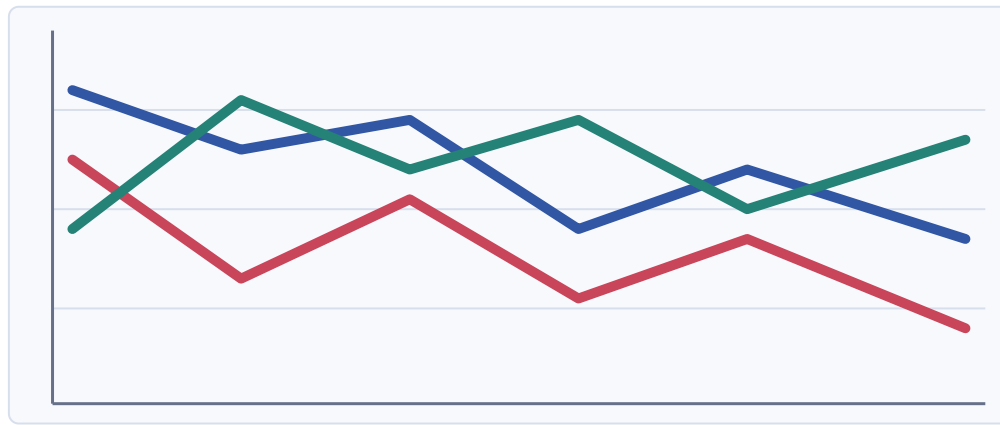
Discriminability is the capacity of a visual channel to represent distinct values that viewers can reliably tell apart. The number and spacing of encoded values must remain within the perceptual capacity of the channel under the viewing conditions and task.

- Three well-separated colors may be easy to name.
- Twenty similar colors may be impossible to distinguish reliably.

Attribute cardinality must fit channel capacity

Fits channel capacity

3 categories encoded by hue

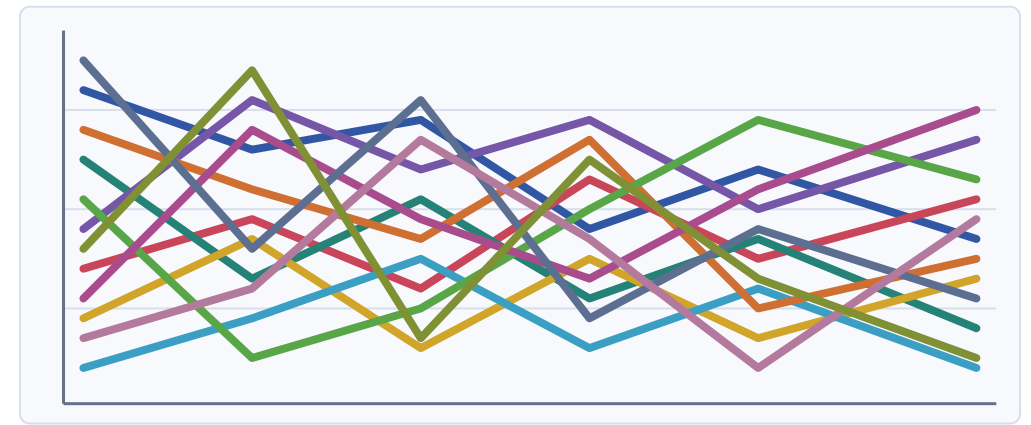


Legend — A — B — C

Three hues remain easy to identify

Exceeds channel capacity

12 categories encoded by hue



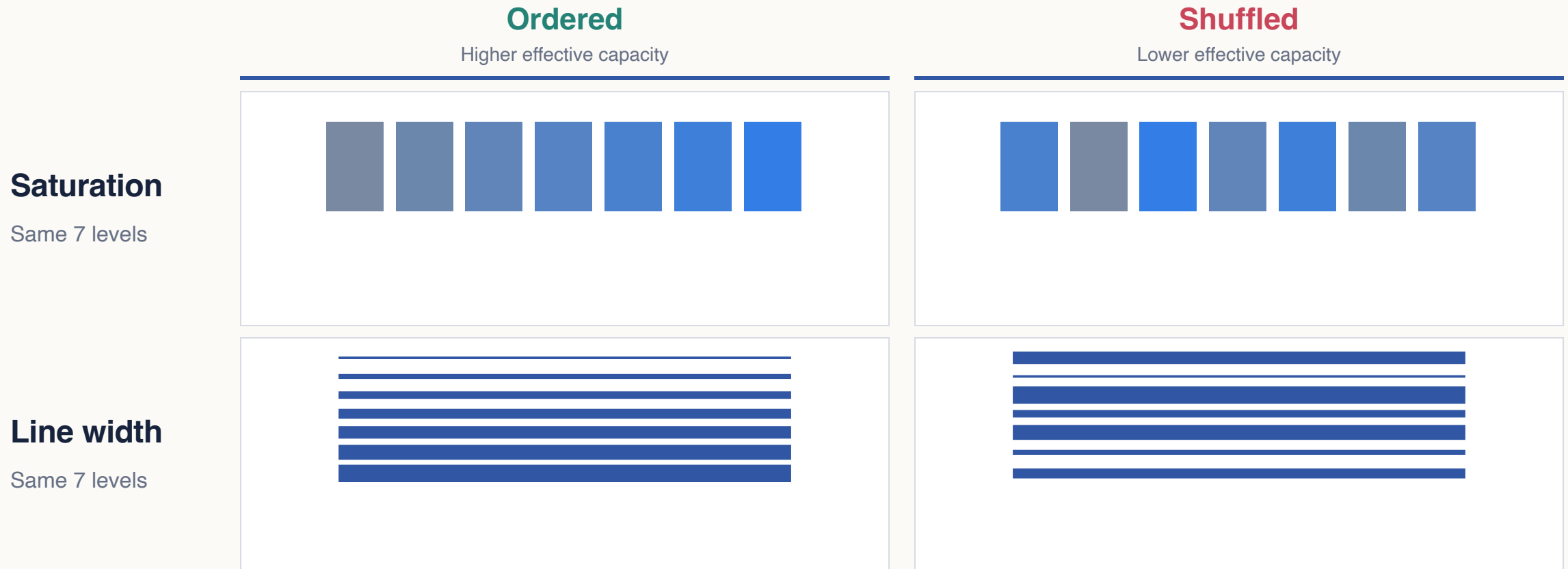
Legend — A — B — C — D — E — F
 — G — H — I — J — K — L

Twelve hues become difficult to track

Discriminability breaks down when attribute cardinality exceeds channel capacity, even if the encoding remains expressive.

Comparison structure shapes effective channel capacity

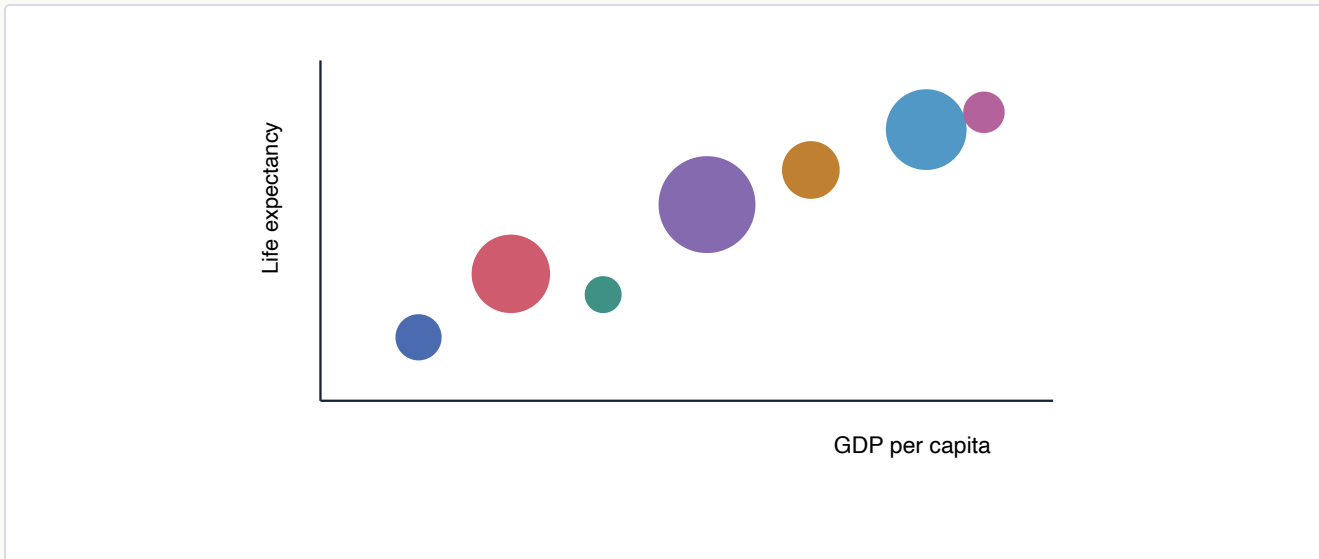
Viewers can have more distinct levels when the layout supports direct comparison.



Arrangement changes comparison difficulty and therefore the effective capacity of the same channel.

Separability describes interference between channels

- Multiple attributes often share one mark, but can viewers read all of them independently?

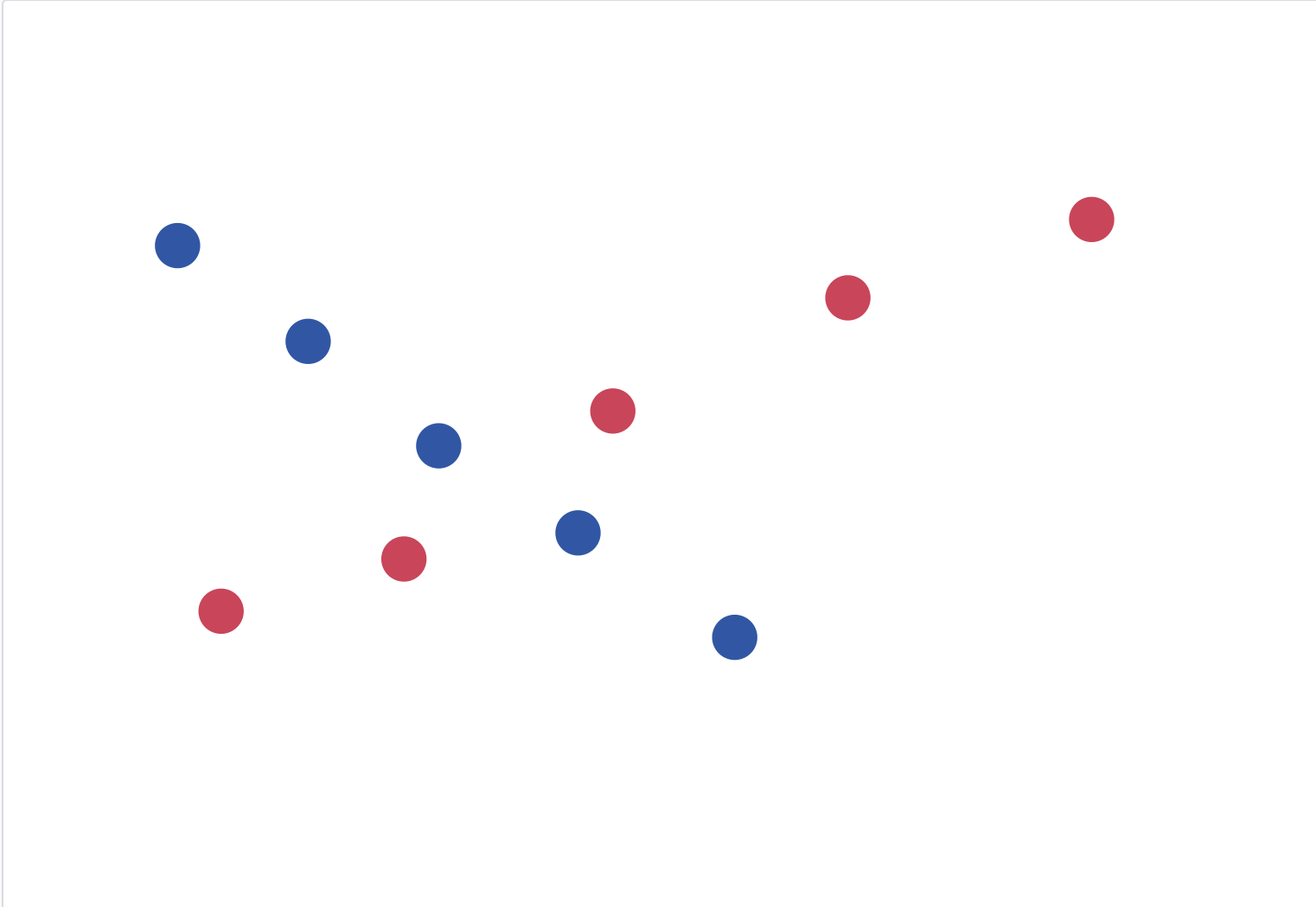


Four mappings

GDP	x position
Life expectancy	y position
Population	size
Continent	hue

Two visual channels are separable when viewers can attend to and interpret the information encoded by one channel without substantial interference from the other.

Position and hue are often separable



Two judgments remain available

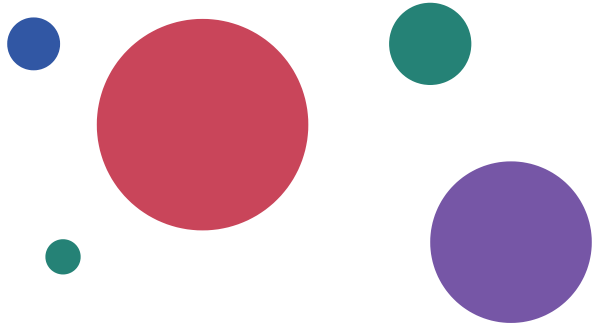
Position reveals the spatial pattern.

Hue reveals category identity.

Viewers can often attend to either with limited interference.

A strong size range can interfere with hue or shape

Size with hue



Large marks attract attention first, so viewers may notice size groups before inspecting the hue categories.

Size with shape



A wide size range can weaken shape recognition because viewers may organize the marks into large and small groups first.

Separability is conditional. Interference increases when one channel creates much stronger visual variation than the other.

Two color mixtures collapse into one perceived color



Attribute A (e.g., GDP) → red intensity

Attribute B (e.g., Population) → blue intensity

Viewers perceive each mixture as a resulting color—not as two independently recoverable hue amounts.

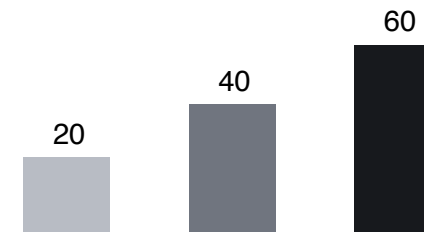
Redundant encoding can strengthen one attribute

Redundant encoding represents the same data attribute using two or more visual channels. It can strengthen an important signal, improve robustness and accessibility, but it also consumes visual capacity and can add unnecessary complexity.

Category → hue + shape

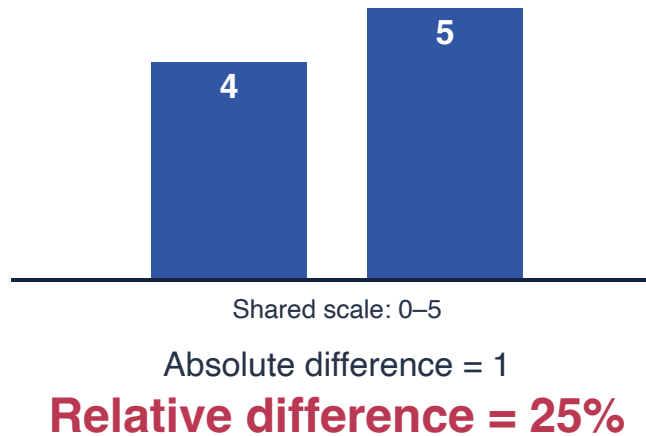
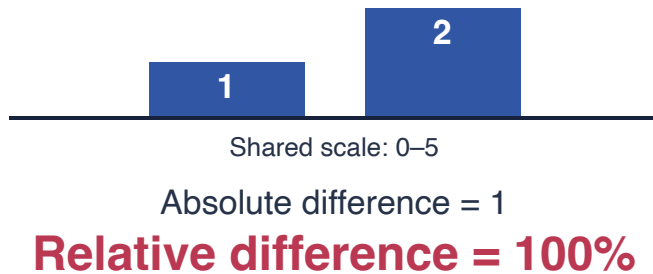


Value → length + luminance



Weber's law: detectable differences are relative

- **Weber's law** states that the smallest detectable change in a stimulus is approximately proportional to the magnitude of the reference stimulus.



$$\Delta I = kI$$

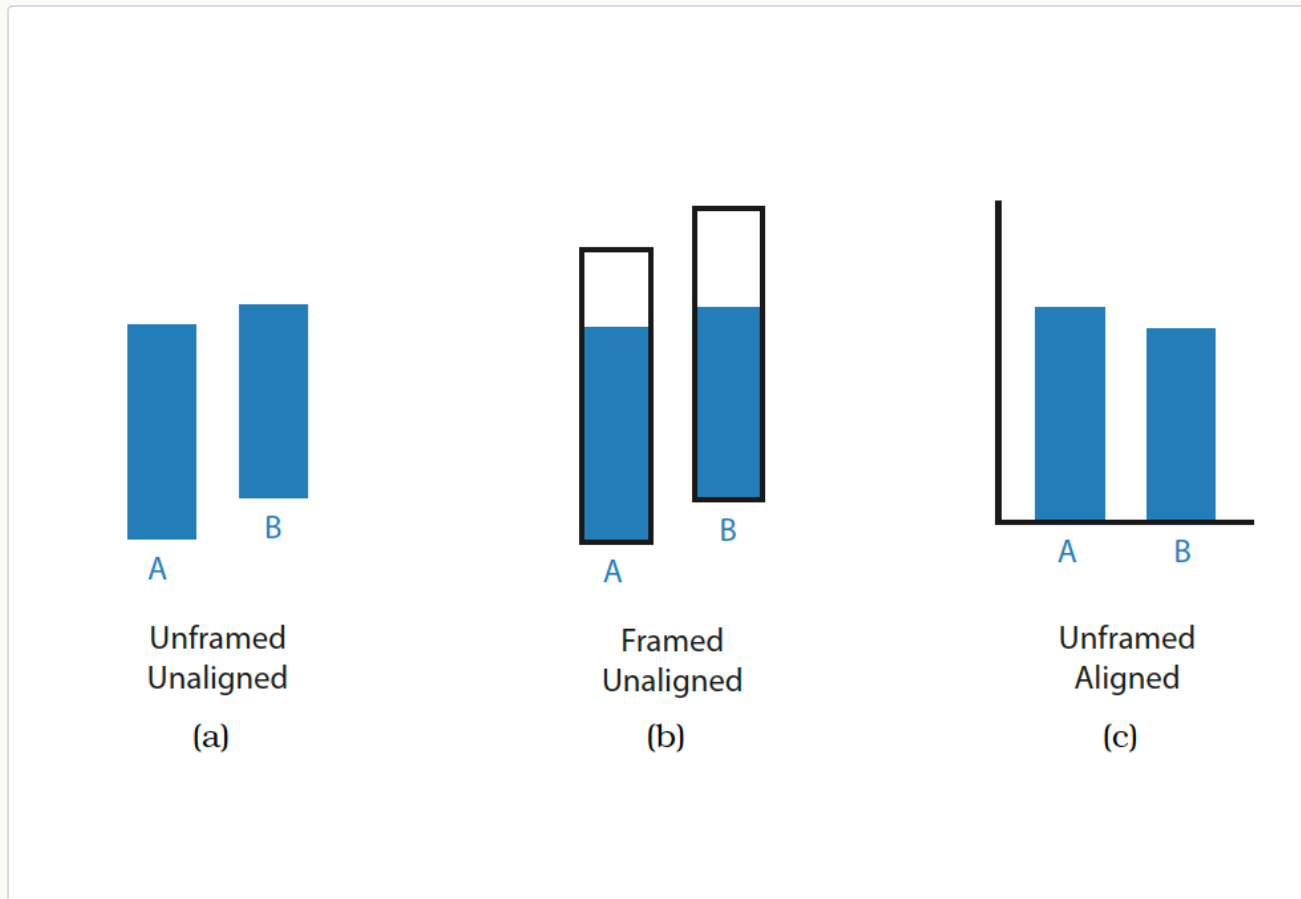
The detectable change grows with the reference intensity.

I	reference intensity
ΔI	smallest detectable change
k	approximate proportion for a sensory condition

Equivalent form: $\Delta I / I \approx k$

Shared references make small differences easier to compare

Weber's law indicates that the human perception is fundamentally based on **relative judgement**, not absolute ones.

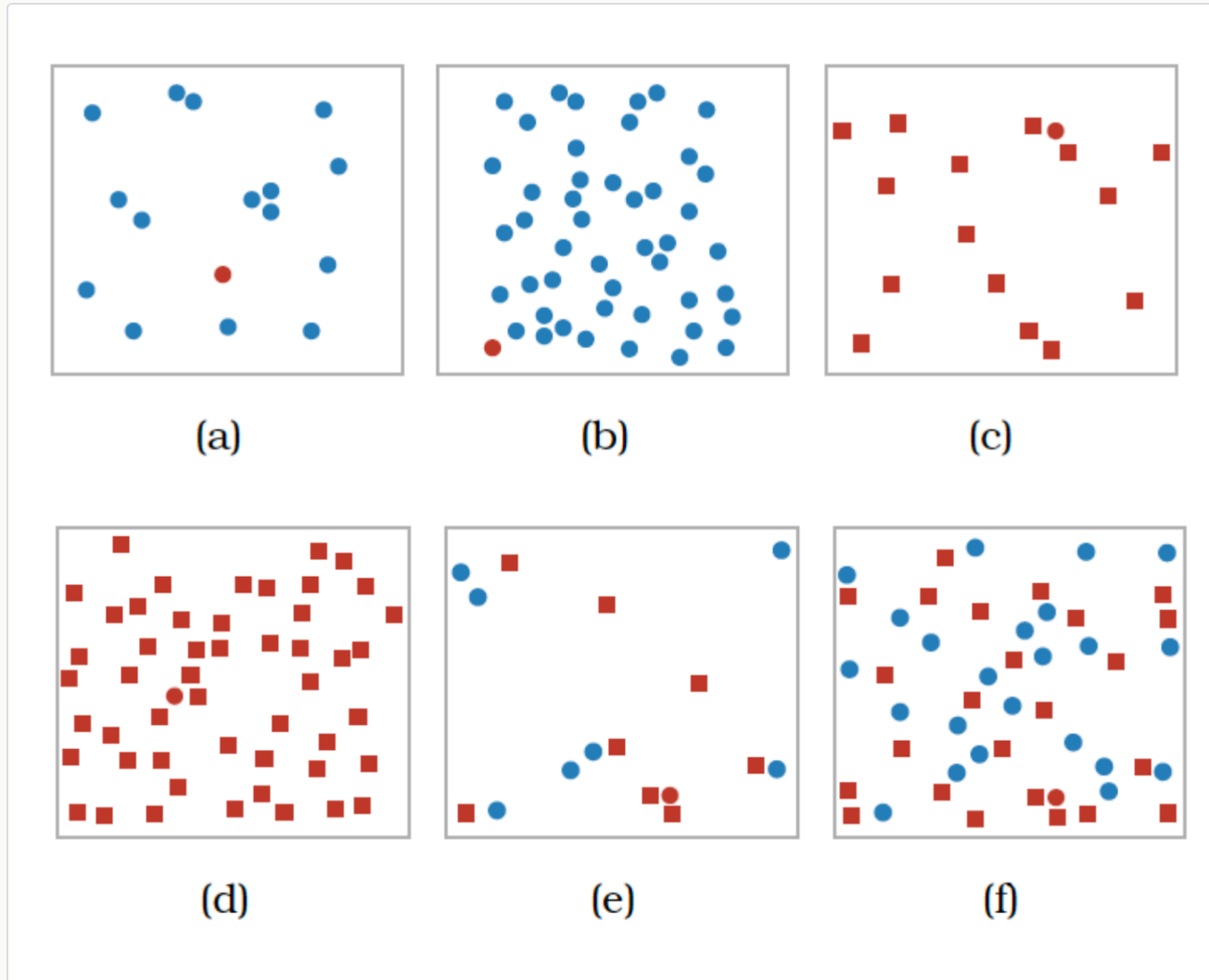


What changes across the three designs?

- **Unframed, unaligned**: viewers should align two isolated lengths mentally.
- **Framed, unaligned**: frames add reference boundaries, but comparison still crosses separate regions.
- **Unframed, aligned**: a common baseline makes the difference directly visible.

Design implication: a stable visual reference reduces comparison effort.

Pop-out depends on contrast with surrounding marks



- From **(a) to (f)**, finding the red target generally takes longer as it shares more visual features with the surrounding distractors and the display becomes denser.
- The greater an object's **visual difference from its surroundings**, the stronger its pop-out effect.

If everything stands out, nothing stands out

Disciplined emphasis

Reference series

Comparison series

Critical exception

Supporting context

Competing emphasis

Revenue

Costs

Growth

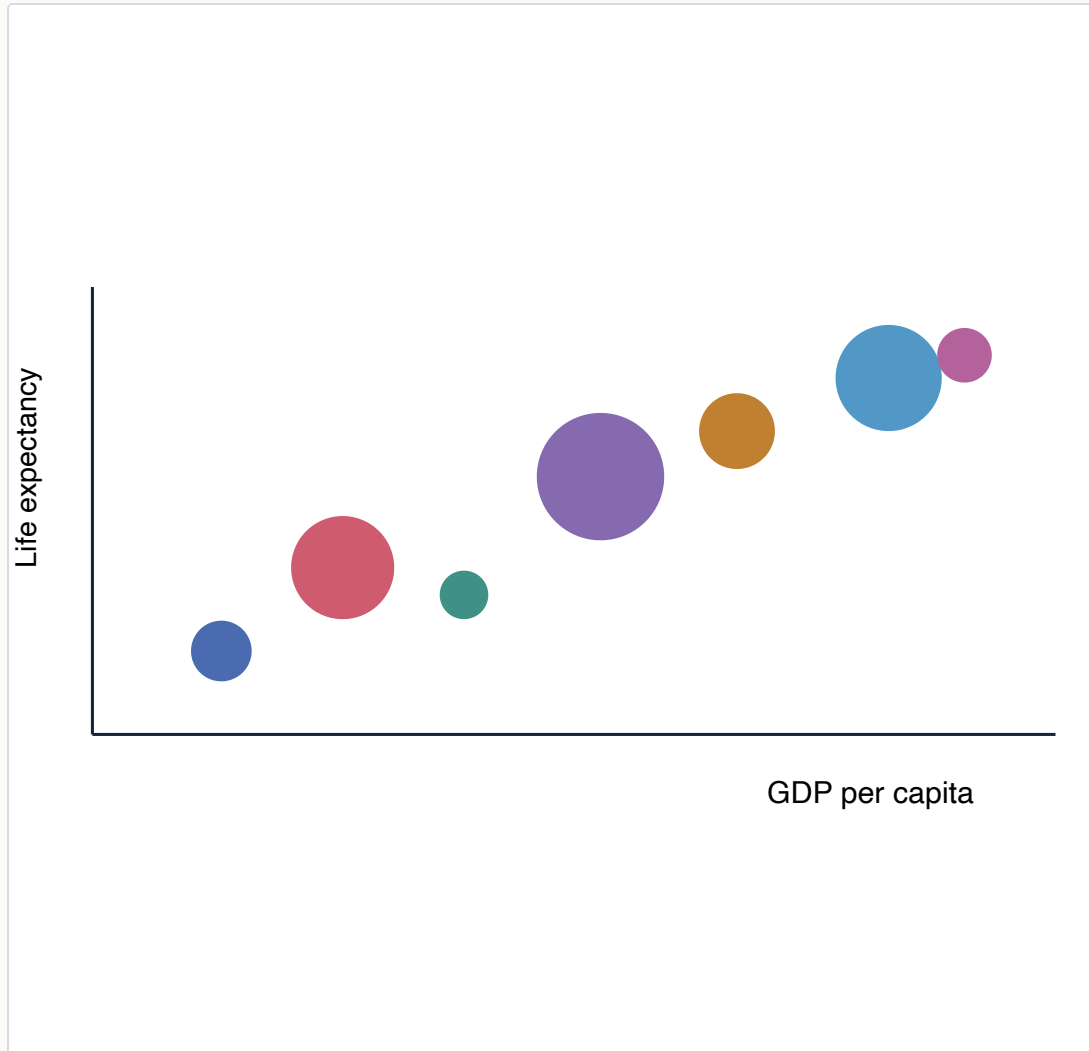
Risk

Target

Forecast

Visual emphasis works only when most elements remain visually quiet.

Evaluate the encoding in the context of its task



TASK

Examine the relationship between GDP and life expectancy across countries.

Expressiveness

GDP and life expectancy are quantitative attributes. Both use ordered position channels.

Effectiveness

The two attributes central to the task receive x and y position, supporting accurate comparison of the overall pattern.

Secondary encodings

Population uses area and continent uses hue. They add context, but the primary task does not require them.

More encoded attributes add information—and cost

Evaluate the added channels

Discriminability	Can viewers distinguish the continent hues?
Visibility	Do large population circles obscure other points?
Separability	Does size interfere with hue?
Effectiveness	Does the task require population?

The cost of another encoding

+ information

but also

+ visual complexity

+ channel competition

+ perceptual load

Encode what matters for the task—not everything available in the dataset.

From a possible encoding to a good encoding

Expressiveness	Does the encoding preserve the meaning and relevant relationships of the data?
Effectiveness	Do task-important attributes receive perceptually strong, suitable channels?
Discriminability	Can viewers reliably distinguish the required values or categories?
Separability	Can multiple channels be read with limited interference?
Attention	Does emphasis direct viewers toward what matters without creating competition?

Lecture 3 constructs an encoding. Lecture 4 evaluates and improves it.